Investment Recommendation System: Project Report

SkillOrbit Investment Banking with Finance Capstone · Oct 17, 2026 · @Rushil Krishna Sai

1. Introduction and objectives

The Investment Recommendation System ranks 28 Nifty 50 stocks by risk and return and labels each one Buy, Hold or Sell, with a one-line reason. It was built for the SkillOrbit Investment Banking with Finance capstone, to show how financial analytics supports investment decisions.

The system meets the five objectives in the project brief:

- Analyse investment data: five years of daily prices for 28 stocks and the Nifty 50 index.
- Compare financial assets: return, risk and trend measures on one common scale.
- Evaluate risk and return: volatility, maximum drawdown, beta and Sharpe ratio, combined into a 0 to 100 risk score.
- Generate recommendations: a transparent scoring rule that produces Buy, Hold or Sell.
- Visualise trends and portfolio insights: a five-page Streamlit dashboard.

Scope. The brief allows stocks, mutual funds or portfolios. This project covers stocks, plus equal-weight portfolios built from them. It does not include real-time trading, advanced quantitative models or brokerage integration, which the brief rules out.

Tools. Python (pandas, NumPy, Matplotlib, Plotly) for the analysis, Streamlit for the dashboard, and Streamlit Cloud for deployment. The code is on GitHub and the dashboard is live at the deployment link.

2. Data

The analysis uses 1,233 trading days of daily prices, from 18 October 2021 to 16 October 2026, for 28 stocks and the Nifty 50 index. The data comes from Yahoo Finance, downloaded with the yfinance Python library, and the prices are adjusted for splits and dividends.

The 28 stocks cover eight sectors:

Sector	Stocks	Count
IT	TCS, Infosys, Wipro, HCL Technologies	4
Banking	HDFC Bank, ICICI Bank, SBI, Axis Bank, Kotak Mahindra Bank	5
FMCG	Hindustan Unilever, ITC, Nestle India	3
Pharma	Sun Pharma, Dr Reddy's, Cipla	3
Auto	Maruti Suzuki, Mahindra & Mahindra, Bajaj Auto	3
Energy	Reliance, ONGC, NTPC, Power Grid	4
Metals	Tata Steel, Hindalco	2
Infra and others	Larsen & Toubro, Asian Paints, UltraTech Cement, Bharti Airtel	4

The Nifty 50 index is downloaded alongside them as the market benchmark.

Cleaning. The raw files had no missing values, no duplicate dates and no price jumps caused by stock splits. The largest one-day move was about 17%. Two problems were fixed:

- **Placeholder rows.** Yahoo had inserted 57 fake rows across the stocks on days with no recorded trading, such as 15 January 2026. Each had zero volume and identical open, high, low and close prices, and each was removed.
- **Misaligned dates.** The index and the stocks did not trade on exactly the same recorded days. Only the 1,233 days common to all 29 series were kept, so every return and beta compares the same days.

3. Methodology

Each stock is measured on seven indicators, which are combined into a 0 to 100 risk score and a 0 to 100 total score. The total score decides the recommendation.

3.1 Indicators

Indicator	What it shows	How it is calculated
Annual return (CAGR)	Average yearly growth	(last price / first price) ^ (1 / years) - 1, over 4.99 years
Volatility	How much the price swings	Standard deviation of daily returns x square root of 252
Maximum drawdown	Worst fall from a previous peak	Lowest value of price / running peak - 1
Beta	Sensitivity to the Nifty 50	Covariance of stock and index daily returns / variance of index returns
Sharpe ratio	Return earned per unit of risk	(CAGR - 7%) / volatility
Trend	Current price direction	Uptrend if price > 50-day average > 200-day average; Downtrend if the reverse; otherwise Sideways
6-month momentum	Recent price change	Price change over the last 126 trading days

The 7% risk-free rate in the Sharpe ratio is an assumption, close to an Indian government bond yield.

3.2 Risk score

Volatility, maximum drawdown (as a positive number) and beta are each scaled from 0 (lowest of the 28 stocks) to 1 (highest), then weighted and multiplied by 100.

Measure	Weight
Volatility	40%
Maximum drawdown	40%
Beta	20%

A score below 33 is Low risk, 33 to 66 is Medium and above 66 is High. The score is relative to this group of 28 stocks, so High means riskier than the others here, not risky in an absolute sense.

3.3 Total score and recommendation

Component	Weight	How it is scored
Historical performance	35%	Sharpe ratio, scaled 0 to 100 across the stocks
Safety	25%	100 minus the risk score
Current trend	25%	Uptrend 100, Sideways 50, Downtrend 0
Recent momentum	15%	6-month momentum, scaled 0 to 100 across the stocks

A stock with a total score of 60 or more is a **Buy**, below 35 is a **Sell**, and anything between is a **Hold**. Each recommendation comes with a one-line reason built from the stock's Sharpe ratio, risk level and trend.

The weights and thresholds are judgement calls, chosen so that the rule stays simple and easy to explain.

4. Results and findings

Seven stocks earn a Buy, twelve a Hold and nine a Sell, and the Buy list is led by ICICI Bank, Bharti Airtel and Sun Pharma. All figures are as of 16 October 2026.

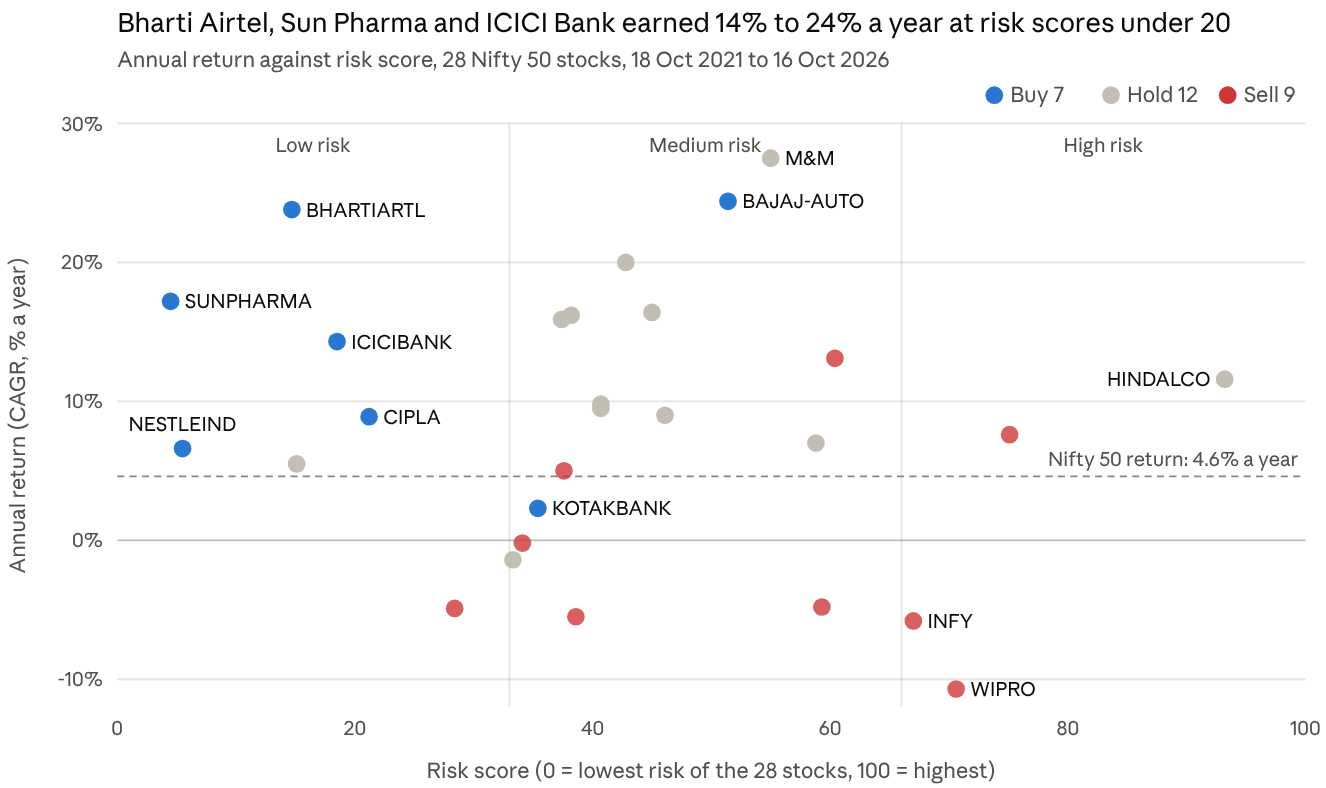
4.1 Recommendations

Buy-rated stock	Sector	Total score	Risk score	Return (CAGR %)	Trend
ICICI Bank	Banking	80.6	18.5	14.3	Uptrend
Bharti Airtel	Infra and others	79.0	14.7	23.8	Sideways
Sun Pharma	Pharma	75.4	4.5	17.2	Sideways
Bajaj Auto	Auto	69.5	51.4	24.4	Sideways
Kotak Mahindra Bank	Banking	67.6	35.4	2.3	Uptrend
Cipla	Pharma	62.0	21.2	8.9	Sideways
Nestle India	FMCG	60.1	5.5	6.6	Sideways

The nine Sell-rated stocks are TCS, ITC, ONGC, Infosys, Reliance, Tata Steel, Asian Paints, Wipro and Hindustan Unilever. Seven of the nine have a negative Sharpe ratio, and most are in a downtrend or carry high risk. The full table for all 28 stocks is in the appendix.

Two cases show how the score balances history against recent price action:

- **Mahindra & Mahindra** has the highest return of all 28 stocks, 27.5% a year, but is only a Hold (50.4) because it is now in a downtrend and has fallen 7.8% in six months.
- **Kotak Mahindra Bank** is a Buy (67.6) despite a 2.3% annual return, because it is in an uptrend and has the best six-month momentum of the group, 16.8%.



Source: Yahoo Finance via yfinance, 18 Oct 2021 to 16 Oct 2026 · scores from the project's recommendation notebook, 28 stocks

Stocks above the dashed line beat the Nifty 50. Bajaj Auto and Mahindra & Mahindra earned the most but carry higher risk, while the Sells cluster at the bottom, at or below zero return.

4.2 Market and trends

The Nifty 50 returned 4.6% a year over the period, with 13.9% volatility. By calendar year it gained 4.3% in 2022, 20.0% in 2023, 8.8% in 2024 and 10.5% in 2025, and has lost 11.4% so far in 2026. It now trades below both its 50-day and 200-day averages.

20 of the 28 stocks beat the index's 4.6% annual return, but each was more volatile than the index on its own: the calmest stock, Nestle India, had 19.6% volatility against the index's 13.9%.

The market is weak at the time of analysis. 13 stocks are in a downtrend, 13 are sideways and only 2 (ICICI Bank and Kotak Mahindra Bank) are in an uptrend. 21 of the 28 stocks have fallen over the last six months. Even long-term strong performers are slipping: NTPC, Power Grid and ONGC are each down 19% to 22% in six months.

4.3 Sectors

Sector	Stocks	Avg return (CAGR %)	Avg risk score	Avg total score
Pharma	3	10.5	13.6	64.8
Banking	5	8.2	33.2	57.6
Auto	3	20.5	49.0	52.3
Infra and others	4	10.9	36.1	46.4
FMCG	3	2.2	23.8	38.9
Energy	4	12.2	43.7	36.4
IT	4	-3.6	63.9	34.2
Metals	2	9.6	84.2	31.6

Pharma scores best, with the lowest average risk. Auto has the highest average return but a middling score, because two of its three stocks are in a downtrend. IT is the weakest sector for returns: TCS, Infosys and Wipro all lost money. Metals is the riskiest, though with only two stocks. Sector averages cover two to five stocks each, so they are indications, not firm conclusions.

4.4 Portfolios

An equal-weight portfolio puts the same amount in every stock it holds.

Portfolio	Return (CAGR %)	Volatility %	Max drawdown %	Sharpe
All 28 stocks	10.3	12.8	-16.6	0.26
7 Low-risk stocks	11.5	12.1	-16.4	0.37
Nifty 50	4.6	13.9	-17.0	-0.17

The Low-risk portfolio earned 11.5% a year against 4.6% for the Nifty 50. Part of that gap is built in, because the Low-risk group was picked using the same five years of history. Both portfolios have lower volatility, 12.1% to 12.8%, than any single stock, which is the benefit of diversification.

5. Dashboard

The results are served through a five-page Streamlit dashboard, deployed publicly on Streamlit Cloud at the deployment link.

Page	What it shows
Overview	The number of Buy, Hold and Sell stocks, and a ranked table of all 28 stocks with their reasons
Stock Explorer	One stock's recommendation, scores, return, volatility, drawdown, beta and Sharpe ratio, with a price chart and its 50-day and 200-day averages
Compare Stocks	Up to five stocks on one growth-of-100 chart, with their indicators side by side
Portfolio Builder	An equal-weight portfolio of chosen stocks against the Nifty 50, with its sector mix
Risk vs Return	All 28 stocks on one scatter chart, coloured by recommendation or risk level

Every page carries the note that the output is for education only and is not financial advice. The source code is in the GitHub repository, with five notebooks that follow the workflow from data overview to recommendations.

6. Limitations

The recommendations are a teaching tool, and six limits should be kept in mind when reading them.

- **Survivorship bias.** The 28 stocks are today's Nifty 50 members. Companies that left the index are not in the data, which flatters the results.
- **In-sample results.** The scores and the Low-risk portfolio are judged on the same five years used to build them. They are a demonstration, not a forecast.
- **Dividend mismatch.** Stock prices include dividends, while the Nifty 50 index line is price-only, so part of the stocks' lead over the index is dividend income.
- **Short window.** Five years covers a single market cycle, and the results would change with a different period.
- **Price history only.** Earnings, valuation and news are not used.

- **Judgement calls.** The score weights, thresholds and 7% risk-free rate are assumptions. Risk levels are relative to these 28 stocks.

7. Conclusion and future work

The system turns five years of price history into a ranked, explainable view of 28 stocks. Pharma stocks, ICICI Bank and Bharti Airtel come out strongest, IT and metals weakest, and the market itself is in a weak phase at the time of analysis. A simple scoring rule gives each stock a Buy, Hold or Sell with a reason a reader can check.

Four extensions would make it stronger:

- Add mutual funds, which the brief lists as an option.
- Add company fundamentals, such as earnings growth and valuation, next to the price-based score.
- Test the recommendations out of sample, by scoring on earlier years and checking how the picks did afterwards.
- Let dashboard users change the score weights and see the ranking update.

Appendix: all 28 stocks

Stocks are listed by NSE ticker and ranked by total score, as of 16 October 2026.

Stock	Sector	Return (CAGR %)	Volatility %	Max drawdown %	Beta	Sharpe	Risk score	Trend	Total score	Recommendation
ICICIBANK	Banking	14.3	20.3	-22.3	0.97	0.36	18.5	Uptrend	80.6	Buy
BHARTIARTL	Infra/Others	23.8	21.6	-18.8	0.80	0.78	14.7	Sideways	79.0	Buy
SUNPHARMA	Pharma	17.2	20.0	-19.1	0.56	0.51	4.5	Sideways	75.4	Buy
BAJAJ-AUTO	Auto	24.4	24.6	-42.3	0.80	0.71	51.4	Sideways	69.5	Buy
KOTAKBANK	Banking	2.3	22.9	-29.9	0.96	-0.21	35.4	Uptrend	67.6	Buy
CIPLA	Pharma	8.9	22.8	-28.3	0.43	0.08	21.2	Sideways	62.0	Buy
NESTLEIND	FMCG	6.6	19.6	-22.0	0.51	-0.02	5.5	Sideways	60.1	Buy
DRREDDY	Pharma	5.5	21.7	-24.2	0.53	-0.07	15.1	Sideways	57.1	Hold
AXISBANK	Banking	9.8	24.4	-28.0	1.08	0.11	40.7	Sideways	52.8	Hold
M&M	Auto	27.5	27.9	-28.1	1.20	0.73	55.0	Downtrend	50.4	Hold
HCLTECH	IT	7.0	25.3	-44.8	0.89	0.00	58.8	Sideways	47.4	Hold
SBIN	Banking	16.2	24.5	-23.9	1.16	0.37	38.2	Downtrend	45.5	Hold
NTPC	Energy	20.0	24.8	-32.3	0.90	0.53	42.8	Downtrend	44.3	Hold
LT	Infra/Others	16.4	24.5	-28.9	1.20	0.38	45.0	Downtrend	43.5	Hold

Stock	Sector	Return (CAGR %)	Volatility %	Max drawdown %	Beta	Sharpe	Risk score	Trend	Total score	Recommendation
HDFCBANK	Banking	-1.4	21.1	-31.0	1.06	-0.40	33.3	Sideways	41.6	Hold
POWERGRID	Energy	15.9	25.1	-29.6	0.76	0.36	37.4	Downtrend	40.9	Hold
ULTRACEMCO	Infra/Others	9.0	23.2	-37.0	1.03	0.09	46.1	Downtrend	38.0	Hold
HINDALCO	Metals	11.6	32.2	-49.9	1.18	0.14	93.2	Sideways	37.6	Hold
MARUTI	Auto	9.5	23.4	-34.7	0.88	0.11	40.7	Downtrend	36.9	Hold
TCS	IT	-4.8	22.9	-53.4	0.82	-0.52	59.3	Sideways	34.3	Sell
ITC	FMCG	5.0	20.2	-45.0	0.67	-0.10	37.6	Downtrend	33.5	Sell
ONGC	Energy	13.1	30.1	-35.3	0.77	0.20	60.4	Downtrend	31.1	Sell
INFY	IT	-5.8	26.2	-48.2	0.96	-0.49	67.0	Sideways	30.0	Sell
RELIANCE	Energy	-0.2	22.4	-27.2	1.11	-0.32	34.1	Downtrend	29.5	Sell
TATASTEEL	Metals	7.6	30.2	-37.4	1.30	0.02	75.1	Downtrend	25.5	Sell
ASIANPAINT	Infra/Others	-5.5	22.2	-39.0	0.73	-0.56	38.6	Downtrend	25.1	Sell
WIPRO	IT	-10.7	25.6	-52.4	0.99	-0.69	70.6	Sideways	24.9	Sell
HINDUNILVR	FMCG	-4.9	20.7	-37.4	0.58	-0.58	28.4	Downtrend	23.2	Sell